Kamale Goutham

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.17
Sri Chaitanya Junior College	Class XII (BIEAP)	2023	90.80%
Sri Chaitanya High School	Class X (BSEAP)	2021	100.00%

SKILLS

Programming Languages: Python, C++, SQL, C, MATLAB, JavaScript

Web: HTML, CSS, JavaScript

Core CS: Data Structures & Algorithms, Operating Systems, DBMS, OOPs

Machine Learning: NumPy, Pandas, Scikit-learn, Matplotlib, XGBoost

Tools: Git/GitHub, LaTeX

RELEVANT COURSES

Computer Science Courses: Introduction to Programming, Data Structures and Applications, Predictive Modelling Introduction to AI/ML, Linear Optimization, Pattern Recognition and Machine Learning, DBMS-1.

Maths Courses: Calculus, Complex Variables, Differential Equations, Vector Calculus, Probability and Random Processes, Matrix Theory, Statistical Inference.

PROJECTS

Online Shopping Web Application | HTML, CSS, JavaScript

- Developed a responsive e-commerce web application using **HTML, CSS, and JavaScript** with product browsing, search, carousel, and cart functionality.
- Implemented **client-side cart management** with item addition/removal, quantity tracking, persistent storage, and checkout flow.
- Designed a responsive **UI/UX** with dynamic product filtering and interactive shopping components.

Online Payment Fraud Detection Using ML | *Python, Scikit-learn, XGBoost, Pandas, NumPy*

- Built an **end-to-end fraud detection pipeline** involving data preprocessing, feature selection, and class-imbalance handling to identify fraudulent online payment transactions.
- Compared Logistic Regression**,** Random Forest, and XGBoost, implementing **class weighting and SMOTE** to improve minority-class detection and evaluate precision-recall trade-offs.
- **Optimized the fraud probability threshold** using a Precision-Recall curve, selecting a balanced operating point that reduced false positives while retaining strong fraud detection performance.

Algorithmic Trading & Backtesting Engine | C++, STL, OOP

- Built a modular **C++ backtesting engine** using historical OHLCV data and pluggable trading strategies.
- Implemented **portfolio simulation** with transaction costs, slippage, position tracking, and trade logging.
- Added **performance analytics** including returns, Sharpe ratio, maximum drawdown, and strategy comparison.

Adaptive Noise Canceller | Prof. Sundaram Vanka

- Developed a MATLAB-based noise-canceller using the **NLMS** algorithm, which actively suppresses background noise in real time based on full or partial suppression mode.

- Used **notch filters** for tone retention and good SNR value, displayed FFT-based plots to evaluate perfomance.
- Implemented real-time simulation and tuning of filter parameters, demonstrating practical understanding of adaptive signal processing concepts.

POSITIONS OF RESPONSIBILITY

- **Informals Coordinator, Elan & nVision**

Organized Nexus with 75+ teams, introduced Mystery Rooms and sponsored wooden games, and guided ~25 deputy coordinators as Informals Coordinator.

- **Hospitality Coordinator, Milan**

Coordinated hospitality for Milan by assisting with guest accommodation, catering, and on-ground support to ensure a smooth experience for participants and dignitaries.

- **Sunshine Mentor, IIT Hyderabad**

Mentored a group of junior students by providing academic guidance, personal support, and organizing sessions to help them adapt smoothly to campus life.

EXTRACURRICULAR

- **Represented IIT Hyderabad** as an 800m long distance runner at **Inter IIT Sports** in 2024.
- **Sports:** Won Gold in Inter-Department badminton DIESTA and Silver in the 800m at Inter-Hostel MILAN, actively involved in **athletics, badminton, and basketball**.
- **Cultural Participation:** Participated in Flash Mobs, Milan Dance competition, and other dance events.

SCHOLASTIC ACHIEVEMENTS

- Secured **AIR 4880** in **JEE Advanced 2023** among 1.8 lakh candidates.
- Placed in the **top 0.95%** among **1.1 million** unique candidates in **JEE MAINS 2023**.
- Achieved a **TS EAMCET 2023** rank of **872** out of approximately 3.2 lakh candidates, placing in the top 0.27% of the state.